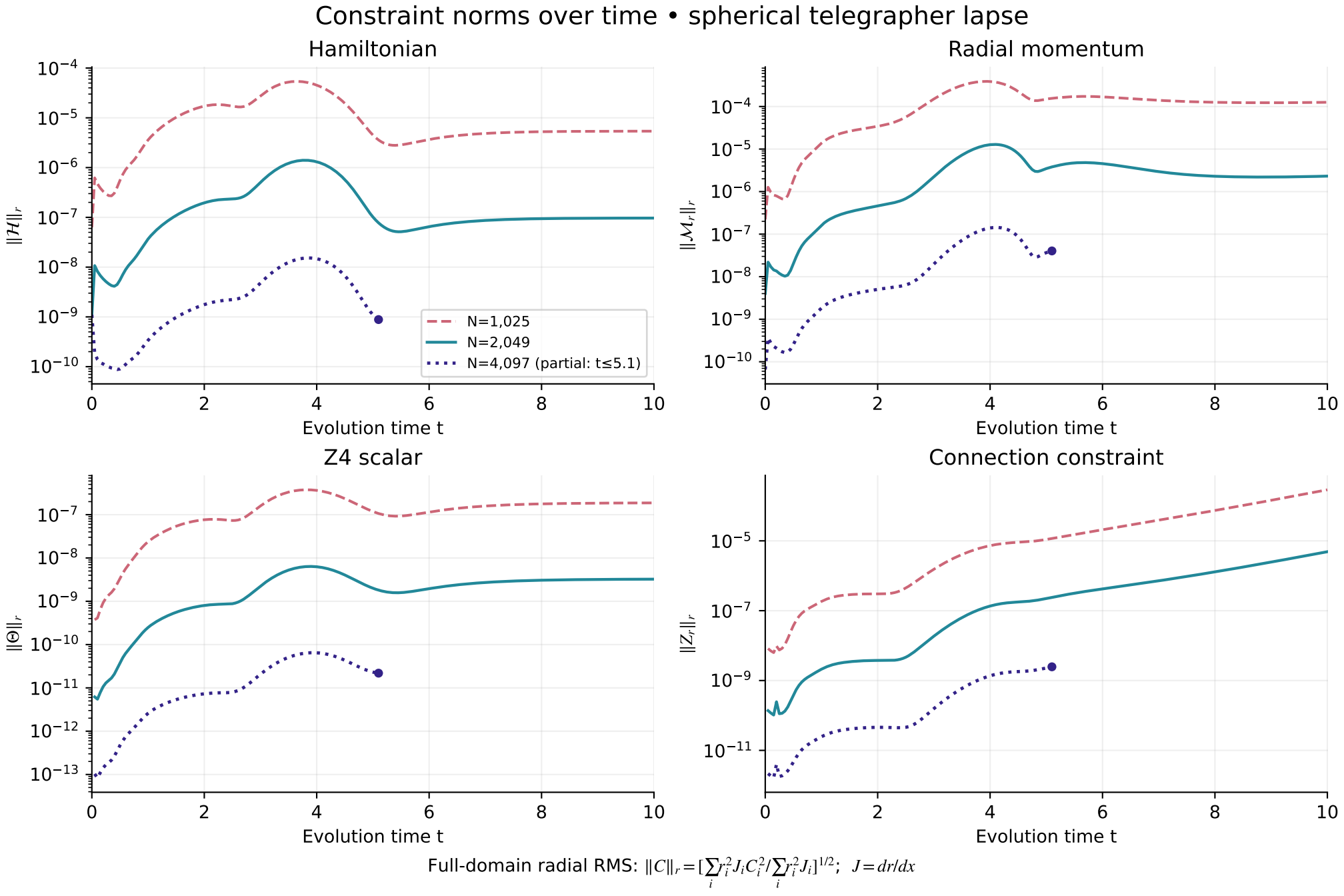
Constraint norms over time • spherical telegrapher lapse
Hamiltonian
Radial momentum
Z4 scalar
Connection constraint
$\|\mathcal{H}\|_r$
$\|\mathcal{M}_r\|_r$
$\|\Theta\|_r$
$\|Z_r\|_r$
Evolution time t
N=1,025
N=2,049
N=4,097 (partial: t≤5.1)
Full-domain radial RMS: $\|C\|_r = [\sum_i r_i^2 J_i C_i^2 / \sum_i r_i^2 J_i]^{1/2}$; $J = dr/dx$